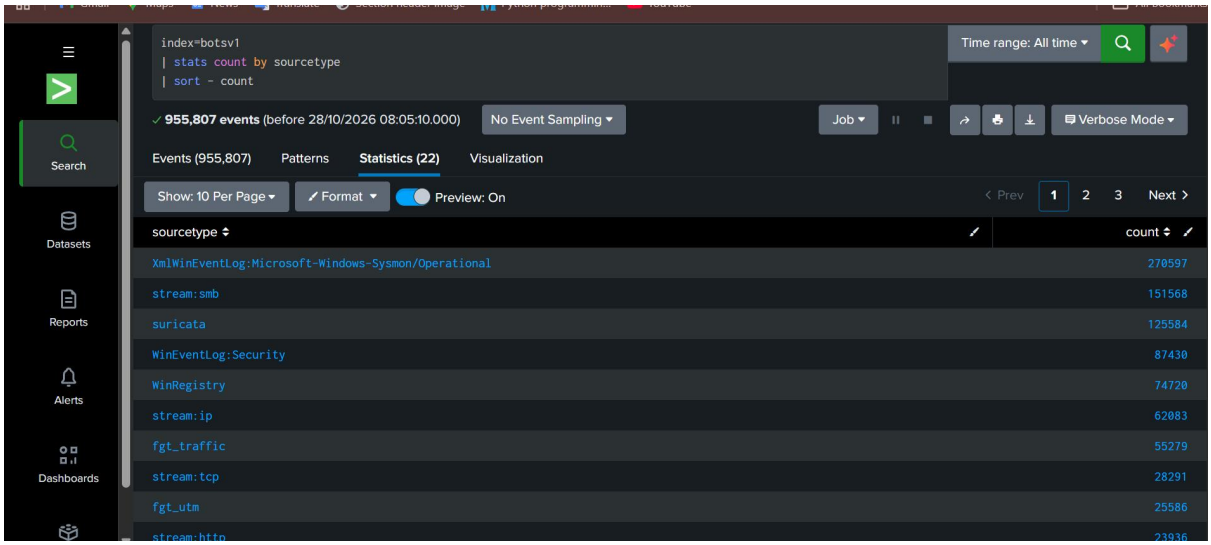
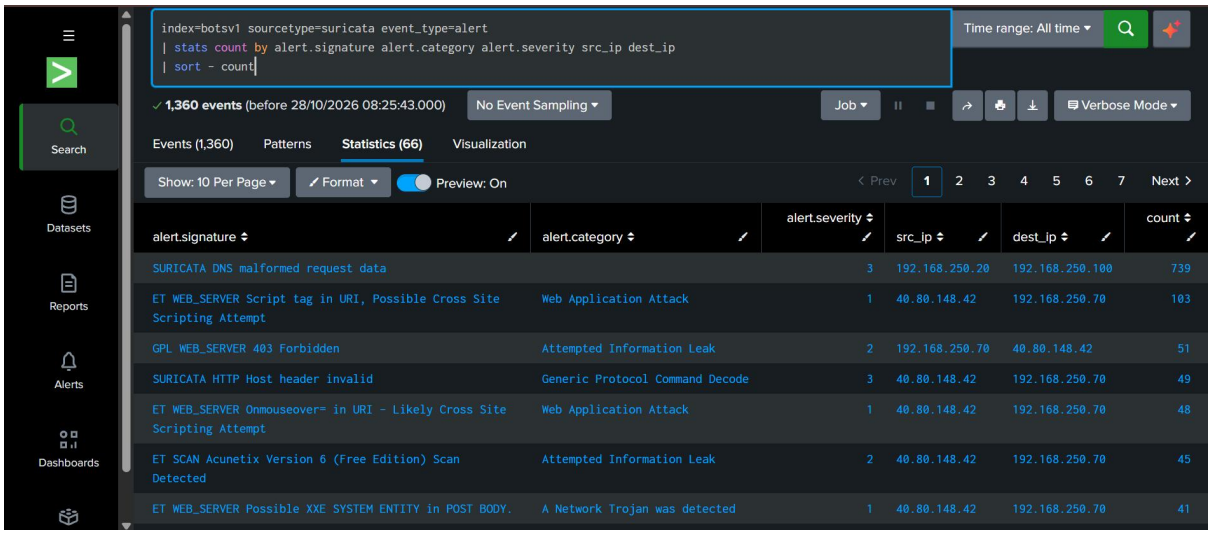
Incident Investigation Using the Splunk BOTS Dataset

Step 1: Scope the dataset



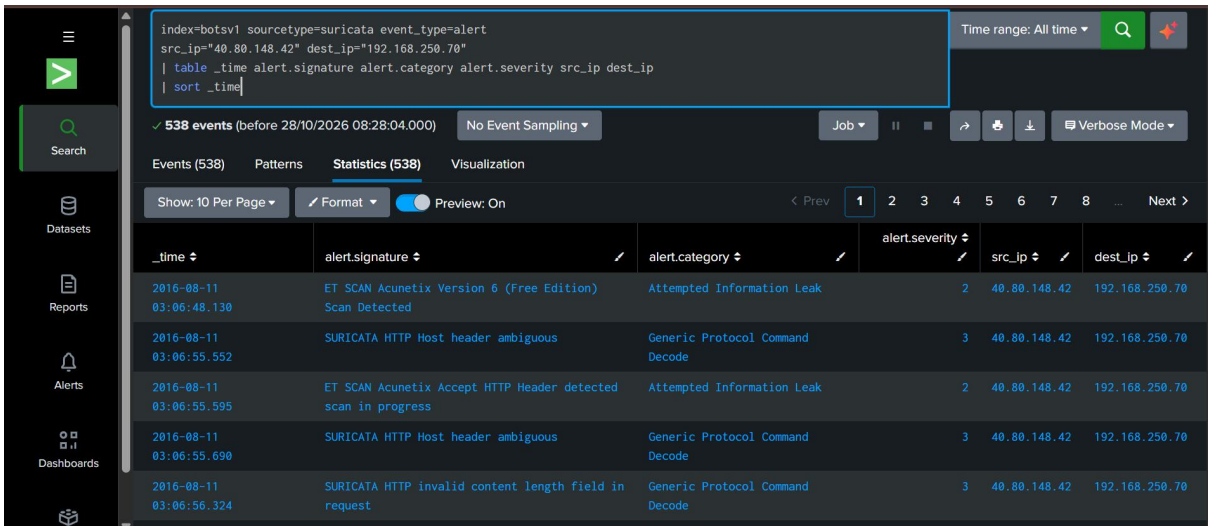
I counted events by sourcetype to see which log sources were available. The index contains 955,807 events across 22 sourcetypes, including Sysmon, Suricata, IIS, stream:http, and FortiGate logs.

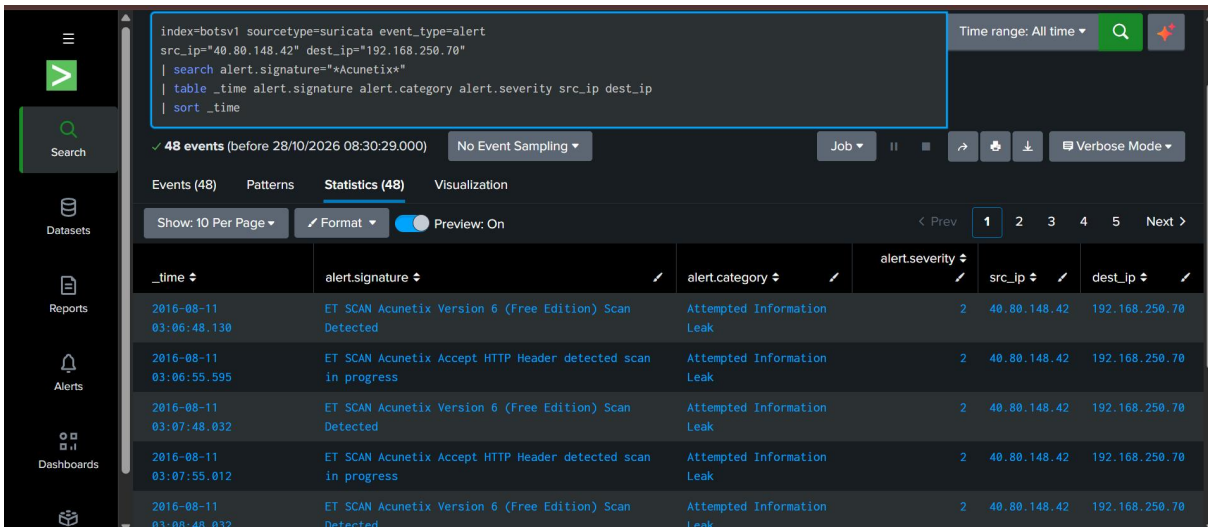
Step 2: Identify high-priority alerts



I searched Suricata alerts grouped by signature, category, severity, and IP pair. Most web-attack alerts, including XSS (103 and 48 events) and XXE (41 events), came from 40.80.148.42 to 192.168.250.70. The largest alert, “SURICATA DNS malformed request data” (739 events), was between internal hosts and was not investigated further.

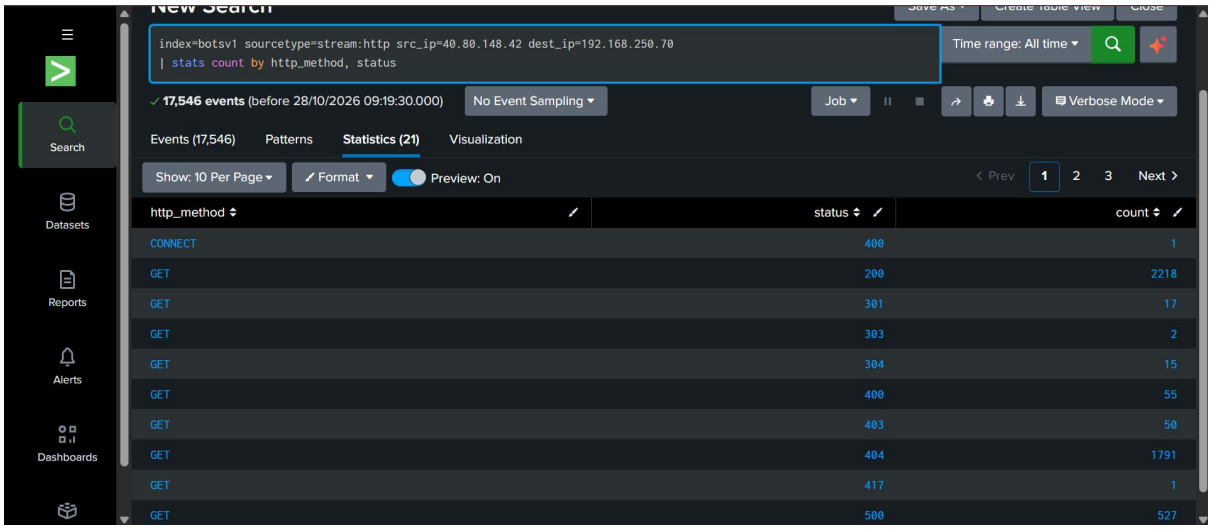
Step 3: Investigate the suspicious IP





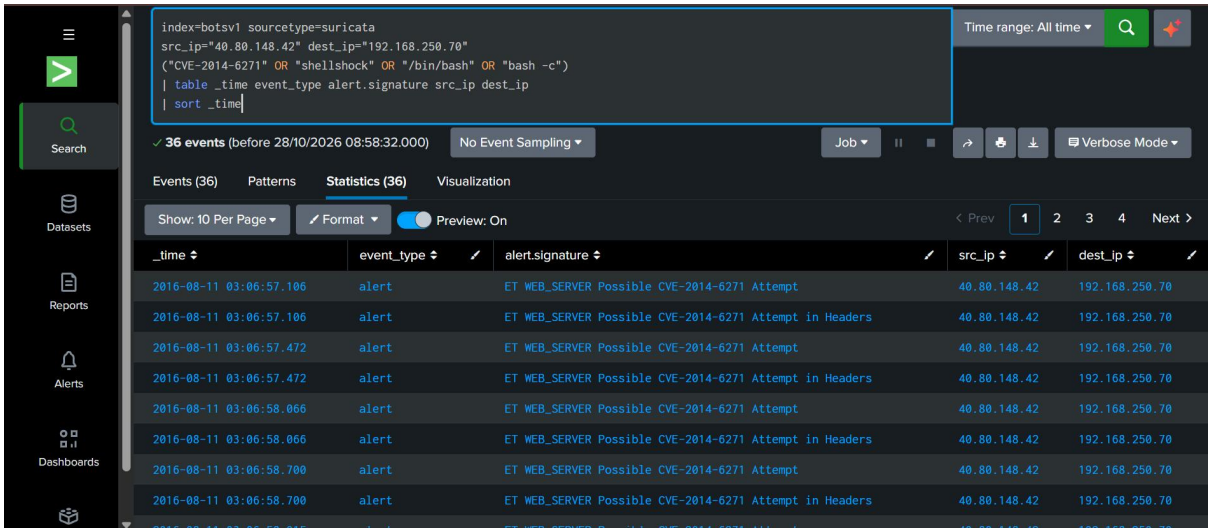
I filtered Suricata alerts for 40.80.148.42 → 192.168.250.70, which returned 538 events search and Acunetrix. 48 of them are Acunetix scanner signatures, showing that this IP ran an automated vulnerability scan against the web server.

Step 4: Check server responses



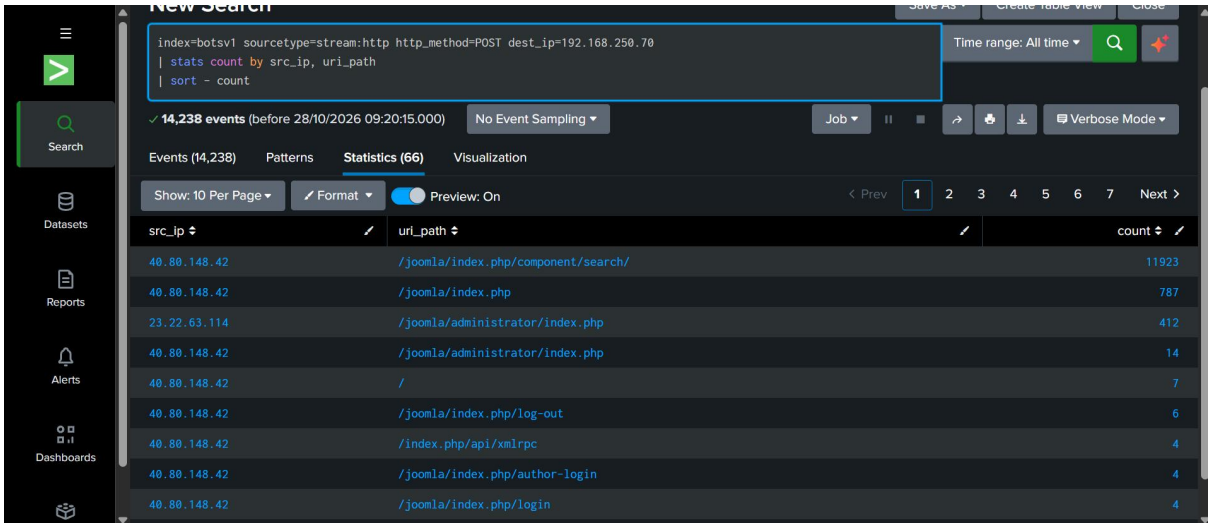
I reviewed how the web server responded to this IP across 17,546 HTTP events. The server returned status 200 (2,218 events), 404 (1,791), and 500 (527), a pattern consistent with automated scanning. A 200 response alone does not prove a compromise.

Step 5: Check Shellshock exploit attempts



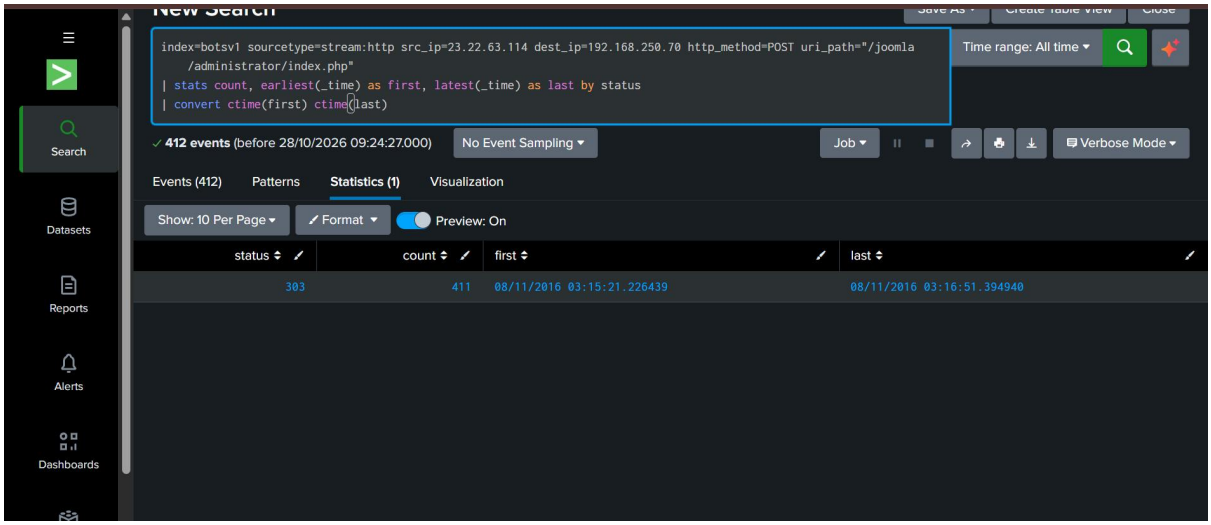
I searched for CVE-2014-6271 (Shellshock) signatures, which returned 36 events starting at 03:06:57. These alerts show exploit attempts only; they do not confirm that any attempt succeeded.

Step 6: Review POST requests



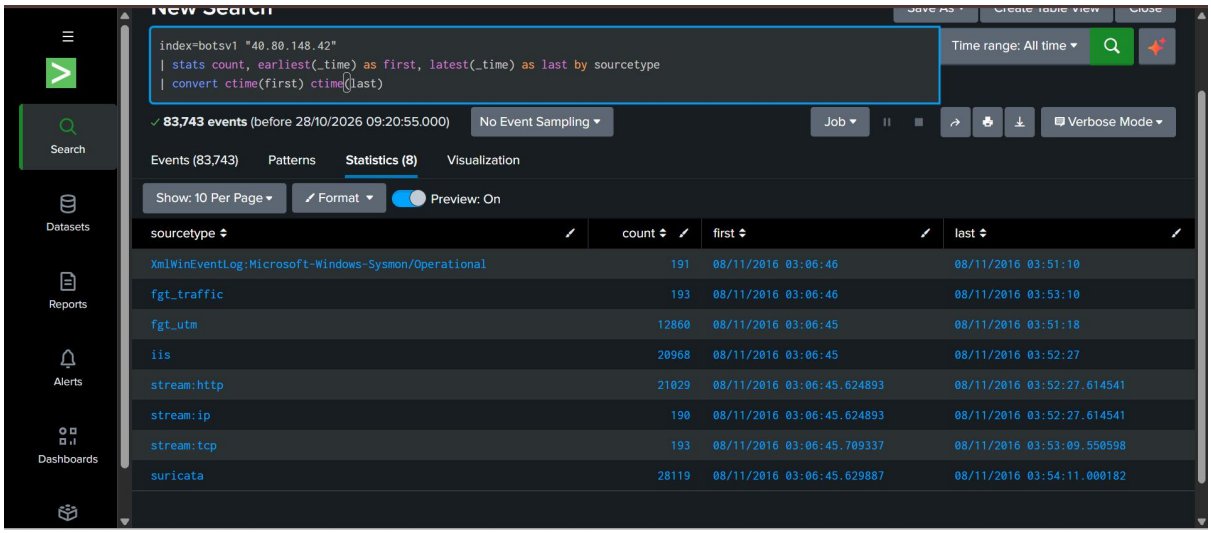
I searched for POST requests to the web server, which returned 14,238 events. 40.80.148.42 sent most of them, mainly to the Joomla search form (11,923) and /joomla/index.php (787). A second IP, 23.22.63.114, sent 412 POST requests to the Joomla administrator login at /joomla/administrator/index.php.

Step 7: Test for brute force



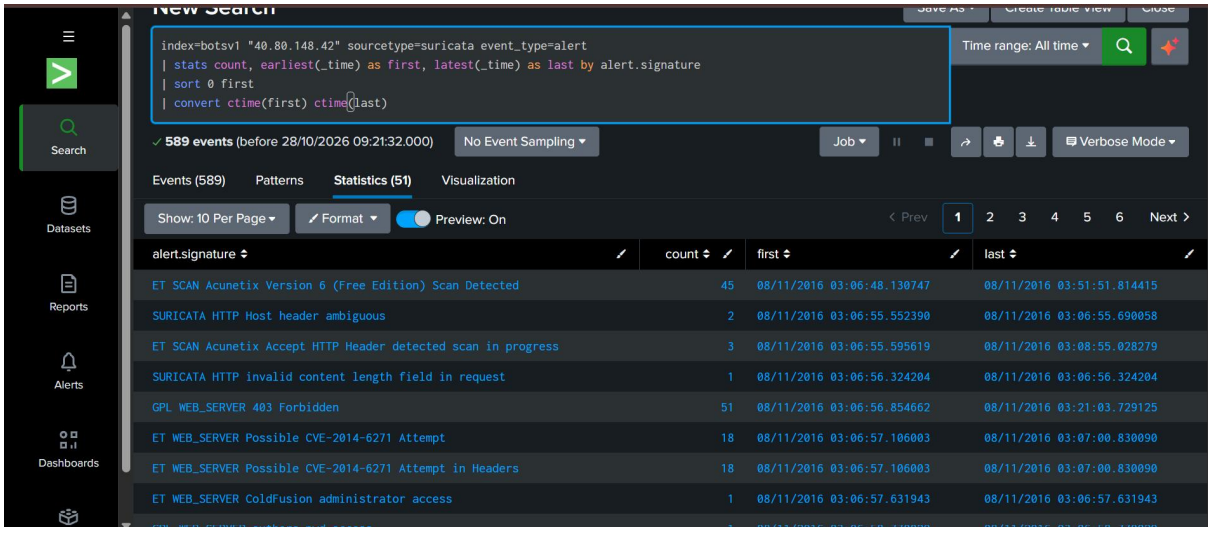
All 412 POST requests from 23.22.63.114 targeted the Joomla admin login between 03:15:21 and 03:16:51, and 411 of them returned the same HTTP 303 response. That speed and uniformity indicate automated brute-force attempts. The logs alone cannot show whether any login succeeded.

Step 8: Correlate across log sources

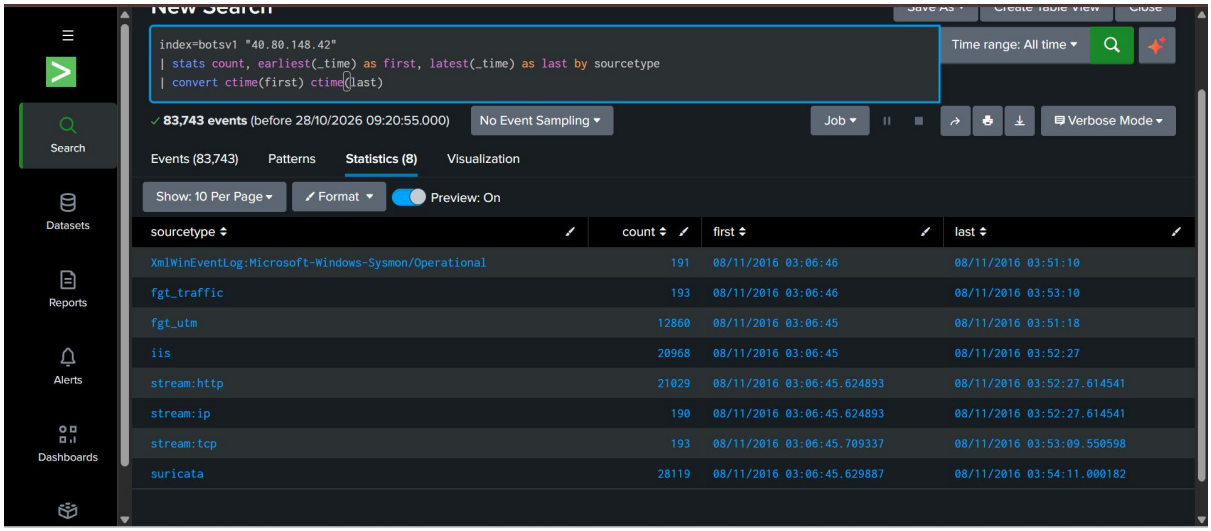


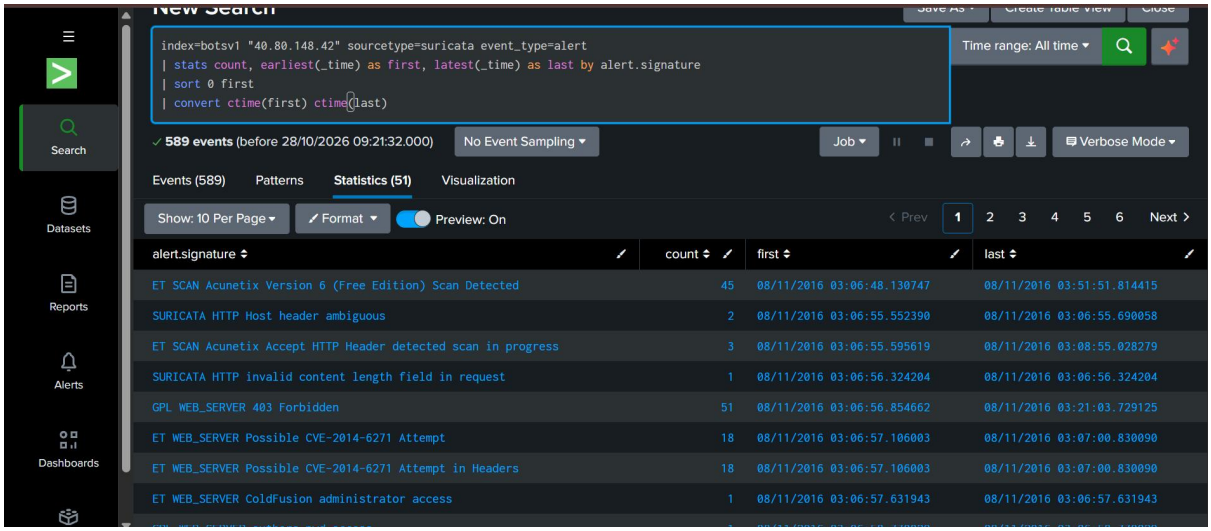
Searching for 40.80.148.42 across the whole index returned 83,743 events in eight sourcetypes: Suricata, stream:http, IIS, FortiGate UTM and traffic, stream:tcp, stream:ip, and Sysmon. All of them fall between 03:06:45 and 03:54:11 on 2016-08-11, so the same activity is visible from the network, web server, and firewall.

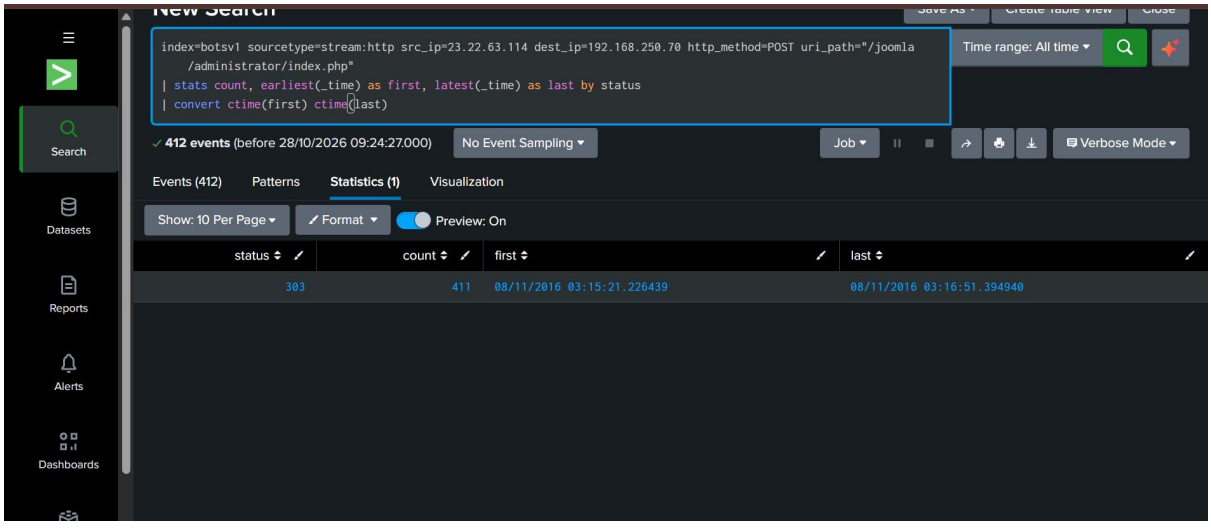
Step 9: Build the alert timeline



I grouped the attacker's Suricata alerts by signature and sorted them by first-seen time. This shows the order of the attack: Acunetix scan, then HTTP anomalies and 403 responses, then Shellshock attempts, all within about twelve seconds of each other.


New Search
index=botsv1 "40.80.148.42"
| stats count, earliest(_time) as first, latest(_time) as last by sourcetype
| convert ctime(first) ctime(last)
Time range: All time
83,743 events (before 28/10/2026 09:20:55.000)
No Event Sampling
Job
Verbose Mode
Events (83,743)
Patterns
Statistics (8)
Visualization
Show: 10 Per Page
Format
Preview: On
sourcetype
count
first
last
XmlWinEventLog:Microsoft-Windows-Sysmon/Operational 191 08/11/2016 03:06:46 08/11/2016 03:51:10
fgt_traffic 193 08/11/2016 03:06:46 08/11/2016 03:53:10
fgt_utm 12860 08/11/2016 03:06:45 08/11/2016 03:51:18
iis 20968 08/11/2016 03:06:45 08/11/2016 03:52:27
stream:http 21029 08/11/2016 03:06:45.624893 08/11/2016 03:52:27.614541
stream:ip 190 08/11/2016 03:06:45.624893 08/11/2016 03:52:27.614541
stream:tcp 193 08/11/2016 03:06:45.709337 08/11/2016 03:53:09.550598
suricata 28119 08/11/2016 03:06:45.629887 08/11/2016 03:54:11.000182
Search
Datasets
Reports
Alerts
Dashboards


New Search
index=botsv1 "40.80.148.42" sourcetype=suricata event_type=alert
| stats count, earliest(_time) as first, latest(_time) as last by alert.signature
| sort 0 first
| convert ctime(first) ctime(last)
Time range: All time
589 events (before 28/10/2026 09:21:32.000)
No Event Sampling
Job
Verbose Mode
Events (589)
Patterns
Statistics (51)
Visualization
Show: 10 Per Page
Format
Preview: On
Prev 1 2 3 4 5 6 Next
alert.signature
count
first
last
ET SCAN Acunetix Version 6 (Free Edition) Scan Detected 45 08/11/2016 03:06:48.130747 08/11/2016 03:51:51.814415
SURICATA HTTP Host header ambiguous 2 08/11/2016 03:06:55.552390 08/11/2016 03:06:55.690058
ET SCAN Acunetix Accept HTTP Header detected scan in progress 3 08/11/2016 03:06:55.595619 08/11/2016 03:08:55.028279
SURICATA HTTP invalid content length field in request 1 08/11/2016 03:06:56.324204 08/11/2016 03:06:56.324204
GPL WEB_SERVER 403 Forbidden 51 08/11/2016 03:06:56.854662 08/11/2016 03:21:03.729125
ET WEB_SERVER Possible CVE-2014-6271 Attempt 18 08/11/2016 03:06:57.106003 08/11/2016 03:07:00.830090
ET WEB_SERVER Possible CVE-2014-6271 Attempt in Headers 18 08/11/2016 03:06:57.106003 08/11/2016 03:07:00.830090
ET WEB_SERVER ColdFusion administrator access 1 08/11/2016 03:06:57.631943 08/11/2016 03:06:57.631943
Search
Datasets
Reports
Alerts
Dashboards


New Search
index=botsv1 sourcetype=stream:http src_ip=23.22.63.114 dest_ip=192.168.250.70 http_method=POST uri_path="/joomla
/administrator/index.php"
| stats count, earliest(_time) as first, latest(_time) as last by status
| convert ctime(first) ctime(last)
Time range: All time
412 events (before 28/10/2026 09:24:27.000)
No Event Sampling
Job
Verbose Mode
Events (412)
Patterns
Statistics (1)
Visualization
Show: 10 Per Page
Format
Preview: On
status
count
first
last
303 411 08/11/2016 03:15:21.226439 08/11/2016 03:16:51.394940
Search
Datasets
Reports
Alerts
Dashboards

Final Findings

On 2016-08-11, between 03:06 and 03:54, the host 40.80.148.42 targeted the web server 192.168.250.70 (imreallynotbatman.com). Suricata detected an Acunetix Version 6 scan from 03:06:48, followed nine seconds later by alerts for CVE-2014-6271 exploit attempts, plus XSS and XXE alerts. Activity from this IP appears in eight log sources. Separately, 23.22.63.114 sent 412 POST requests to the Joomla administrator login between 03:15:21 and 03:16:51, indicating an automated brute-force attempt.

Conclusion: Scanning, exploit attempts, and brute-force attempts are confirmed. A successful compromise is **not confirmed** by the evidence reviewed.

Incident Timeline (2016-08-11)

Time	Event	Log source	Meaning
03:06:45	First requests from 40.80.148.42	stream:http, IIS	Scanning begins
03:06:48	Acunetix Version 6 scan detected	Suricata	Scanner identified
03:06:56	First “403 Forbidden” alert (51 total)	Suricata	Server blocking requests
03:06:57	CVE-2014-6271 attempts begin (36 alerts)	Suricata	Shellshock exploit attempt
03:15:21–03:16:51	412 POSTs from 23.22.63.114 to Joomla admin login	stream:http	Brute-force attempt
03:51:51	Last Acunetix alert	Suricata	Scan ends
03:54:11	Last Suricata alert from 40.80.148.42	Suricata	Activity ends

Indicators of Compromise (IOCs)

Type	Value	Context
IP	40.80.148.42	Scanning and exploit-attempt source
IP	23.22.63.114	412 POSTs to the Joomla admin login
IP	192.168.250.70	Targeted web server
Hostname	imreallynotbatman.com	Target site
Tool	Acunetix Version 6	Scanner
URI	/joomla/administrator/index.php	Login target
CVE	CVE-2014-6271	Shellshock exploit attempts

Limitations

A successful compromise was not confirmed. 40.80.148.42 also sent 14 POST requests to the Joomla login, and Sysmon events mentioning it were not analysed.